

COMP-6915: Machine Learning

Project Proposal

Exoplanet Detection from Stellar Light Curves Using Machine Learning

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Introduction

The identification of exoplanets is a fundamental scientific pursuit, providing insight into planetary formation, stellar evolution, and potential habitability. Missions such as Kepler and TESS have produced extensive photometric datasets enabling large-scale study of stellar brightness variations [Borucki et al., 2010, Ricker et al., 2015]. Machine learning methods have demonstrated strong performance in automating transit detection and reducing manual workload in astrophysical data analysis [Shallue & Vanderburg, 2018].

Problem Statement

This project aims to develop a machine learning model that classifies whether a stellar light curve contains evidence of a planetary transit. Transit signals are characterized by periodic decreases in flux but are often obscured by stellar variability, instrumental noise, or systematic artifacts. The challenge is to distinguish genuine transits from false positives effectively and reliably.

Dataset

The project will utilize the Kepler Exoplanet Search dataset, available through sources such as Kaggle or the NASA Exoplanet Archive. These datasets contain labeled light curves representing confirmed planets, false positives, and ambiguous cases. Preprocessing includes detrending, phase folding, smoothing, and normalization. Periodograms may also be computed to identify periodic structures.

Methodology

Initial models include Logistic Regression and Random Forests using engineered statistical features. Deep learning models, such as 1D Convolutional Neural Networks (1D-CNNs), have proven effective in detecting transit patterns due to their ability to learn local temporal features [Pearson et al., 2018]. Recurrent neural networks (LSTM/GRU) will be considered for modeling sequential structure. Model selection and tuning will follow established machine learning best practices.

Evaluation

Models will be evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. Data will be divided into training, validation, and test subsets using stratified sampling when appropriate. Comparative evaluation will assess the relative strengths of feature-based and deep learning methods in transit classification.

Expected Outcomes

The expected outcome is a robust classification model capable of detecting exoplanet transit signatures with high reliability. Beyond predictive performance, the project aims to highlight the role of machine learning in accelerating scientific discovery and enabling scalable analysis of astronomical datasets.

References

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